Can Hospitals Efficiently Integrate Health Care Financing and Delivery?

Evidence from Provider-Sponsored Medicare Advantage Plans

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Abstract

Some hospitals integrate health care financing and delivery by sponsoring their own health plans. We study how this integration affects hospital quality, measured by 30-day unplanned readmissions and 30-day mortality, in Medicare Advantage market. Integration changes hospitals' incentives by allowing the health system to internalize the plan-side savings from reducing downstream utilization, thereby strengthening incentives to invest in quality. Exploiting within-hospital changes in plan-sponsorship status over time and quasi-random variation in hospital assignment generated by ambulance referral patterns, we find little evidence of an average effect of plan sponsorship on readmission and mortality rate in the full sample. In unconcentrated hospital markets, however, admission to a plan-sponsoring hospital reduces the probability of a 30-day unplanned readmission by 12.4 percentage points and the probability of 30-day mortality by 5.6 percentage points. The results suggest that hospital sponsorship of a health plan can lead hospitals to improve patient outcomes, but that the benefits vary strongly across markets with different levels of competition.

Keywords: Hospital Readmission, Mortality, Provider-Sponsored Health Plans, Vertical Integration, Medicare Advantage.

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1 Introduction

Provider-sponsored health plans (PSHPs) are health plans sponsored by healthcare providers and represent a form of integration between healthcare financing (the plan side) and healthcare delivery (the provider side). Provider-sponsored health plans have become increasingly prevalent among U.S. hospitals. The number of hospitals owning or jointly owning a provider-sponsored health plan increased from 692 (18.3% of hospitals) in 2018 to 834 (27.2%) in 2023 (Chen et al., 2026). Proponents argue that PSHPs can better align incentives between payers and providers, facilitate care coordination, and potentially improve efficiency and quality (Frakt et al., 2013; Meyers et al., 2020). Critics emphasize that provider–insurer integration may also create incentives to steer patients toward affiliated providers, weaken competition, and increase market power, with gains from integration therefore not necessarily translating into better patient outcomes (Cuesta et al., 2024; Lin et al., 2021; Bejarano et al., 2024a).

This paper provides new empirical evidence on whether hospital plan sponsorship influences inpatient quality, using 30-day unplanned readmissions and 30-day mortality among Medicare Advantage (MA) beneficiaries. Because MA plans receive capitated payments from Medicare, plan sponsorship exposes the health system to the downstream utilization of beneficiaries enrolled in its plan and can therefore strengthen incentives to reduce avoidable hospital use. Under conventional arm's-length arrangements, plans and hospitals each bear some of the costs of readmissions, and insurers can use negotiated rates and contract terms to encourage hospitals to prevent them. For patients enrolled in a hospital-sponsored plan, the sponsoring health system also internalizes the plan-side savings from improving post-discharge care and reducing readmission risk. Plan sponsorship therefore creates stronger incentives to reduce avoidable readmissions among the system's own enrollees.

We refer to hospitals that sponsor a plan as vertically integrated (VI) hospitals. VI hospitals may respond to the financial incentives in two ways. First, they may make costly upfront investments in quality and care-management capacity—for example, hiring nurses and case managers, improving discharge planning, medication reconciliation, post-discharge monitoring, and care-coordination protocols—that reduce the expected probability and cost of subsequent readmissions. Because many such investments involve hospital-wide protocols and organizational changes, their effects may extend beyond patients enrolled in the hospital's own plan. Second, hospitals may engage in steering patients to an outpatient setting which would lower the cost without improving patient outcome. We therefore examine 30-day mortality as a complementary outcome that is less directly affected by hospital steering.

We address two conceptually distinct sources of endogeneity. The first is supply-side

selection: hospitals with different VI status may also differ along other dimensions that affect readmission risk. We address time-invariant differences using within-hospital changes in VI status across the two years in our data.

The second concern is patient-side selection: the composition of beneficiaries admitted to a hospital may change when the hospital's VI status changes. If healthier patients are systematically more likely to choose VI hospitals, but this sorting pattern is constant over time, hospital fixed effects absorb the resulting time-invariant differences in patient composition. However, if the composition of patients changes when a hospital becomes vertically integrated, within-hospital comparisons may still be biased. Because we do not want to impose the assumption that patient sorting remains unchanged around VI adoption, we complement the fixed-effects design with an instrumental-variables strategy. Following the logic of Doyle Jr et al. (2015), we construct an ambulance-company-based instrument that exploits differences in ambulance referral patterns across emergency episodes to generate plausibly exogenous variation in admission to VI hospitals.

A large literature in health economics and industrial organization links hospital market structure to quality and operational performance (Kessler and McClellan, 2000; Gaynor et al., 2015; Cooper et al., 2022; Chandra et al., 2016; Short and Ho, 2020). We therefore examine whether the effect of plan sponsorship differs across hospital markets with different levels of concentration. In the MA analysis, we classify hospitals using county HHI and estimate the model separately for hospitals in unconcentrated markets (HHI below 1,800) and concentrated markets (HHI of at least 1,800). The Traditional Medicare analysis uses an analogous split based on 30-minute-drive HHI. The 1,800 cutoff follows the threshold at which the 2023 DOJ/FTC Merger Guidelines classify a market as highly concentrated.[1] These are heterogeneity analyses: because market concentration is not randomly assigned and the estimates come from separate subsamples, the comparisons do not identify the causal effect of competition.

We construct a pooled sample of two years of administrative data on Medicare Advantage beneficiaries. We observe beneficiaries' plan enrollment and use the AHRQ Compendium of Health Systems to determine whether an MA plan is sponsored by a hospital organization. Claims and encounter data provide patient location, ambulance transportation, the admitting hospital, principal diagnosis, inpatient length of stay, and 30-day outcomes. We estimate linear probability models with hospital-by-diagnosis fixed effects and instrument for hospital assignment using differential distance and ambulance destination patterns. For Traditional

[1]For expositional simplicity, we call the group below 1,800 "unconcentrated." This is a binary label: it includes some markets that earlier agency guidelines described as moderately concentrated. See the 2023 DOJ/FTC Merger Guidelines.

Medicare beneficiaries, we do not observe claims-level records. Instead, we observe aggregate hospital-condition readmission and mortality rates reported as three-year averages. These data provide a longer panel than the MA claims, which we use for two purposes: to assess pre-adoption trends and to test whether the effect of MA plan sponsorship extends beyond MA enrollees through hospital-wide changes in care. We retain four non-overlapping three-year reporting cycles and use the Callaway and Sant'Anna staggered difference-in-differences estimator to compare hospitals before and after qualifying plan sponsorship with hospitals that are never treated during the sample.

In the MA full sample, the IV estimate for readmission is negative but statistically insignificant, while the estimated reduction in 30-day mortality is 3.9 percentage points and is statistically significant at the 10 percent level. When we estimate the same model separately by local markets with different levels of concentration, the estimates in unconcentrated markets imply reductions of 12.4 percentage points in readmission and 5.6 percentage points in mortality. Both are statistically significant. The corresponding estimates in concentrated markets are smaller and statistically insignificant. These IV estimates are local effects for patients whose hospital assignment is shifted by differential distance and ambulance destination patterns, and their large magnitudes warrant caution.

The Traditional Medicare analysis provides complementary evidence. In unconcentrated 30-minute-drive markets, qualifying plan sponsorship is associated with a 0.525 percentage point reduction in the risk-standardized readmission rate. The corresponding readmission estimates in the full sample and in concentrated markets are not statistically significant, and we find no statistically significant overall mortality effect in any market group. Event-study estimates in the unconcentrated-market sample show no individually significant pre-adoption coefficients, although the confidence intervals are wide, and negative readmission effects after adoption. Thus, the MA and Traditional Medicare analyses both point to lower readmissions in their respective unconcentrated-market subsamples, but neither design establishes that market competition causes this pattern.

The paper connects three related literatures. Research on managed care shows that insurer incentives can alter utilization, spending, and patient outcomes (Cutler et al., 2000; Baicker et al., 2013; Afendulis et al., 2017; Curto et al., 2019; Agafiev Macambira et al., 2026). Work on spillovers further shows that managed-care incentives can change provider behavior for patients outside the plan directly generating those incentives (Glied and Zivin, 2002; Chernew et al., 2008; Baicker et al., 2013). Our Traditional Medicare analysis tests for precisely this type of hospital-wide response to MA plan sponsorship.

The paper also relates to evidence on hospitals' responses to quality incentives. Readmissions declined after the Hospital Readmissions Reduction Program (Zuckerman et al., 2016;

Desai et al., 2016), and Gupta (2021) shows that hospitals responded through both improvements in care and changes in how returning patients were classified. We study a different incentive: sponsoring a capitated MA plan changes the health system's return to preventing downstream utilization rather than imposing an explicit penalty for excess readmissions. The debate over the HRRP effects also motivates our use of mortality as a complementary outcome (Ody et al., 2019; Ziedan and Kaestner, 2021).

Finally, a growing literature examines integration between health systems and MA plans. Integrated plans differ in premiums, star ratings, patient experience, medical loss ratios, utilization, and outcomes (Meyers et al., 2020; Park et al., 2023; Bejarano et al., 2024b,a; Ma et al., 2026; Bejarano et al., 2026). This evidence is largely descriptive or cross-sectional. We instead combine within-hospital changes in sponsorship with an IV design for hospital assignment, and use both MA beneficiary-level and Traditional Medicare hospital-level outcomes to study whether plan sponsorship changes the quality of inpatient care.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, the data used in the estimation, and the descriptive statistics. Section 3 discusses the sources of endogeneity and the instrumental variables. Section 4 presents evidence using hospital-level data on Fee-for-Service beneficiaries. Section 5 presents the main empirical setting and shows the evidence using MA beneficiaries claims and enrollment data. Section 6 concludes.

2 Institutional Setting and Data

This section describes the institutional environment of Medicare Advantage (MA), the role of provider-sponsored plans, and the data used in the empirical analysis. We first discuss the growth of vertically integrated provider-sponsored health plans and how these organizations combine payer and provider functions. We then explain why hospital readmissions provide a useful outcome for evaluating the effects of hospital–plan integration. We then present a simple toy model that illustrates the differential incentives of a vertically integrated hospital as compared to a standard hospital and how the incentives differ across market concentration. Finally, we describe the administrative datasets used in the analysis and present descriptive statistics on readmission rates and patient episodes across different integration settings.

2.1 Medicare Advantage and Provider-Sponsored Plans

Medicare Advantage (MA) is the private-plan alternative to Traditional Medicare. Medicare Advantage beneficiaries enroll in health plans offered by private insurance companies. Pri-

vate plans are paid by Medicare a capitated amount per-beneficiary per month to provide coverage for TM Part A and Part B benefits. Plans are differentiated in terms of provider network and coverage, and negotiate with providers the payment rates for each service. Provider-sponsored health plans (PSHPs) are becoming a prominent feature of the Medicare Advantage (MA) market. PSHPs are vertically integrated entities in which a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. Common ownership may align incentives across payer and provider, reducing transaction costs and enabling clinical integration, potentially raising quality and efficiency. Yet integration can also enable patient steering and increased market power. This vertically integrated organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals. In this paper we will exclusively focus on plans sponsored by health systems. We will refer to hospitals that sponsor a plan as vertically integrated (VI) hospitals.

2.2 Hospital Readmission, Mortality, and Hospital–Plan Integration

Hospital readmissions are a clinically and financially meaningful outcome for studying hospital–plan integration. A rehospitalization within 30 days can reflect failures in discharge planning, medication reconciliation, follow-up, or post-acute care management, and readmissions are explicitly targeted by Medicare's Hospital Readmissions Reduction Program. In a conventional arm's-length arrangement, the insurer bears the claims cost of a rehospitalization while the hospital may continue to receive payment for the additional admission. When a hospital system sponsors an MA plan, the organization also captures the plan-side savings from preventing avoidable downstream utilization. Common ownership can therefore strengthen the return to investments in care transitions and post-discharge management.

Many of these investments involve common staff, information systems, and clinical protocols rather than treatment that can easily be limited to the sponsoring plan's enrollees. Improvements in discharge planning or care-management capacity may thus extend to Traditional Medicare beneficiaries and other patients treated at the same hospital. This potential hospital-wide response motivates our complementary analysis of aggregate Traditional Medicare outcomes. It also links the paper to evidence that managed-care incentives can spill over to patients outside the plan directly creating those incentives (Glied and Zivin, 2002; Baicker et al., 2013).

Readmission alone does not fully characterize quality. Measured readmissions may decline if a hospital treats a returning patient in an outpatient or observation setting rather than admitting the patient, even without an improvement in underlying health (Gupta, 2021). We therefore examine 30-day mortality as a complementary outcome that is not mechanically affected by inpatient-versus-outpatient classification. Mortality cannot identify the precise mechanism behind a readmission decline, but it helps assess whether lower measured readmissions are accompanied by worse short-term survival.

2.3 Toy Model

2.3.1 Incentives for Vertically Integrated Hospitals

This paper focuses on 30-day hospital readmissions as a clinically and financially meaningful outcome through which to study hospital–insurer integration. Let R_i denote an indicator equal to one if patient i experiences an unplanned inpatient readmission within 30 days of discharge, and let

$$q_h \equiv \Pr(R_i = 1 \mid h)$$

denote the probability of readmission following treatment at hospital h. Thirty-day readmissions are common and costly among Medicare beneficiaries and are explicitly targeted by Medicare's Hospital Readmissions Reduction Program (HRRP).[2] Because readmissions are sensitive to discharge planning, transitional care, and post-discharge management, they provide a natural outcome for studying whether changes in hospitals' financial incentives translate into changes in care delivery.

To illustrate how plan sponsorship changes these incentives, consider first a non-integrated arrangement in which an MA plan reimburses a hospital for an inpatient hospitalization. Hospital payment contracts in Medicare Advantage can take different forms, but existing evidence suggests that inpatient payments commonly remain closely tied to Traditional Medicare reimbursement rates. Berenson et al. (2015), for example, find that MA plans typically pay hospitals rates approximately equal to Traditional Medicare rates, while Maeda and Nelson (2018) similarly find that MA inpatient hospital payments are roughly equal to Medicare fee-for-service payments on average. Traditional Medicare, in turn, reimburses acute inpatient hospitals prospectively on a per-discharge basis, with payments determined primarily by the patient's MS–DRG.

[2]CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with "excess" readmissions relative to national benchmarks. See Gupta (2021).

Motivated by this evidence, let p_h denote the incremental payment received by hospital h from an insurer when an additional inpatient readmission occurs, and let c_h denote the corresponding treatment cost. The hospital's contribution margin from an additional readmission is

$$m_h = p_h - c_h.$$

If the probability of readmission is q_h, the expected contribution margin associated with readmissions is

$$q_h(p_h - c_h).$$

If $m_h > 0$, an additional readmission generates positive contribution margin for the non-integrated hospital. Absent countervailing incentives such as HRRP penalties or other quality-based payment arrangements, the hospital therefore has no direct financial incentive to reduce readmissions. By contrast, the insurer bears the claims cost associated with the additional hospitalization. When the insurer and hospital are separate organizations, the hospital therefore does not fully internalize the downstream savings generated by preventing rehospitalization.

Plan sponsorship changes this calculation. Let B_i denote the risk-adjusted capitated payment received by an MA plan for beneficiary i. Importantly, B_i does not depend on the cost of care needed to treat patient i. Let $k_h > 0$ denote the cost to the integrated organization associated with an additional readmission. If the readmission occurs at a hospital owned by the same system, k_h reflects the resources required to provide the additional hospitalization. If the readmission occurs at an outside hospital, k_h reflects the payment that the plan must make to the outside provider.

For a beneficiary enrolled in the hospital system's own MA plan, there are two possible states. With probability q_h, the beneficiary is readmitted and the integrated organization receives the capitated payment B_i but incurs the downstream cost k_h. With probability $1 - q_h$, the beneficiary is not readmitted and the organization receives B_i without incurring this additional cost. Expected surplus can therefore be written as

$$\Pi_h^{VI} = q_h(B_i - k_h) + (1 - q_h)B_i.$$

Simplifying,

$$\Pi_h^{VI} = B_i - q_h k_h.$$

Hospitals may respond to this incentive by investing in care-management capacity. Let e_h denote investment in activities such as discharge planning, care-management staffing, timely outpatient follow-up, medication reconciliation, post-discharge monitoring, and coordination with outpatient and post-acute providers. Suppose that investment lowers the probability of readmission,

$$q_h = q(e_h), \qquad q'(e_h) < 0,$$

and let $F(e_h)$ denote the cost of investment. Expected surplus for an own-plan beneficiary can then be written as

$$\Pi_h^{VI}(e_h) = B_i - q(e_h)k_h - F(e_h).$$

Because B_i does not depend on e_h, the marginal return to investment for the hospital system's own-plan beneficiaries is

$$\frac{\partial \Pi_h^{VI}(e_h)}{\partial e_h} = -q'(e_h)k_h - F'(e_h).$$

Thus, plan sponsorship increases the return to investments that reduce readmissions among beneficiaries enrolled in the hospital system's own plan: when investment lowers $q(e_h)$, the integrated organization avoids downstream costs that would otherwise be borne by the insurer.

Importantly, many investments in care management may operate at the level of the hospital or clinical practice rather than being targeted exclusively to patients enrolled in a particular insurance plan. Changes in discharge protocols, care-management staffing, information systems, medication reconciliation, or coordination with post-acute providers may therefore affect the treatment of patients across payers. This possibility is consistent with evidence that managed-care incentives can generate spillovers to patients outside the plans directly subject to those incentives (Glied and Zivin, 2002; Baicker et al., 2013). Plan sponsorship can therefore create incentives originating from the hospital system's own insured population while generating changes in care for a broader set of patients treated at the hospital.

At the same time, such hospital-wide spillovers introduce an additional tradeoff. Preventing a readmission for a patient insured by an outside plan may cause the hospital to forgo the contribution margin that it would otherwise earn from that admission. Whether the financial benefit generated by the hospital's own-plan population is sufficiently large to induce greater hospital-wide investment therefore depends on the composition of patients and on the provider margin associated with admissions of patients insured by outside plans.

We formalize this tradeoff in the next subsection and show how it varies with hospital market concentration.

2.3.2 Hospital Market Concentration and Vertically Integrated Hospitals

Hospital market structure may shape the extent to which vertical integration changes hospitals' incentives to invest in reducing readmissions. The key mechanism arises when investments in care management operate at the hospital level and therefore affect patients enrolled both in the hospital system's own Medicare Advantage plan and in outside insurance plans.

Let α_h denote the share of patients at hospital h who are enrolled in the hospital system's own MA plan. As above, let e_h denote hospital-wide investment in care management and let the probability of readmission be $q(e_h)$, where

$$q'(e_h) < 0.$$

For patients enrolled in the hospital's own MA plan, a readmission creates a downstream cost for the integrated organization. Let $k_h > 0$ denote the cost internalized by the integrated system when one of its own-plan patients is readmitted. Preventing such a readmission therefore generates a financial benefit of k_h to the integrated organization.

The incentives are different for patients insured by outside plans. Let p_h denote the payment received by the hospital from an outside insurer for an additional inpatient admission, and let c_h denote the corresponding treatment cost. The hospital's contribution margin from a readmission of an outside-plan patient is

$$m_h = p_h - c_h.$$

When $m_h > 0$, preventing a readmission for an outside-plan patient causes the hospital to forgo this contribution margin. Hospital-wide investments therefore generate two opposing financial effects for an integrated system: they reduce downstream spending for patients enrolled in the system's own plan, but they may also reduce profitable admissions among patients insured by outside plans.

Let $F(e_h)$ denote the cost of hospital-wide care-management investment. The portion of the integrated organization's expected profit that depends on readmissions and investment can be written as

$$\Pi_h^{VI}(e_h) = -\alpha_h q(e_h) k_h + (1 - \alpha_h) q(e_h) m_h - F(e_h).$$

where α_h denote the share of patients treated at hospital h who are enrolled in the hospital

system's own Medicare Advantage plan. The first term captures the expected downstream cost of readmissions among the hospital system's own plan enrollees. The second captures the contribution margin generated by readmissions among patients insured by outside plans. Differentiating with respect to investment gives

$$\frac{\partial \Pi_h^{VI}}{\partial e_h} = -q'(e_h)\left[\alpha_h k_h - (1-\alpha_h)m_h\right] - F'(e_h).$$

Thus, the net marginal financial benefit from reducing readmissions is

$$\alpha_h k_h - (1-\alpha_h)m_h.$$

The first component, $\alpha_h k_h$, represents the savings generated by preventing downstream utilization among the integrated system's own enrollees. The second component, $(1-\alpha_h)m_h$, represents the provider margin forgone when the same hospital-wide investment prevents readmissions among patients insured by outside plans. A strictly positive level of investment is optimal when the marginal benefit of reducing readmissions is sufficiently large to exceed the marginal cost of investment. In particular, a sufficient condition for positive investment is

$$-q'(0)\left[\alpha_h k_h - (1-\alpha_h)m_h\right] > F'(0).$$

Hospital market concentration affects this tradeoff through the provider-side margin m_h. Hospitals in more concentrated markets have greater bargaining power vis-à-vis insurers and may therefore obtain higher negotiated prices. Holding treatment costs fixed,

$$p_h^C > p_h^U$$

implies

$$m_h^C > m_h^U,$$

where C and U denote concentrated and unconcentrated hospital markets, respectively. Consequently,

$$\alpha_h k_h - (1-\alpha_h)m_h^C < \alpha_h k_h - (1-\alpha_h)m_h^U.$$

The marginal return to hospital-wide investments that reduce readmissions is therefore lower in concentrated hospital markets. Under standard assumptions that investment costs are convex and that the marginal effectiveness of care-management investment diminishes

with investment, and holding α_h, k_h, and the technology $q(e_h)$ fixed across markets, this implies

$$e_h^C < e_h^U.$$

The intuition is that vertical integration creates an incentive to reduce readmissions among the hospital system's own plan enrollees because the system internalizes the resulting downstream savings. However, when the resulting care-management investments also reduce readmissions among patients insured by outside plans, the hospital gives up the contribution margin associated with those admissions. This opportunity cost is larger when negotiated hospital prices are high. Consequently, the additional incentive created by plan sponsorship to make hospital-wide investments in reducing readmissions is stronger in more competitive hospital markets, where the provider-side margin forgone when an admission is prevented is smaller.

This mechanism therefore generates the empirical prediction that the reduction in readmissions associated with hospital plan sponsorship should be larger in unconcentrated hospital markets than in concentrated hospital markets.

2.4 Data

Medicare Advantage Claims Data We use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan.

The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. We exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans.

The main data source for patients' acute-care utilization is the Inpatient Encounter Data. These files identify the admitting hospital (CCN) and, for ambulance-transported episodes, the ambulance provider. They include ICD-10 diagnosis and procedure codes, length of stay, and discharge disposition, which we use to construct episode definitions, risk adjusters, and readmission outcomes.

Provider-sponsored health plans and VI hospitals. To identify integrated MA plans sponsored by health systems, we use the AHRQ Compendium of Health Systems, which contains the list of all health plans offered by hospitals and health systems. We link the hospitals in the Compendium to their CMS provider identification numbers (CCN) in order to merge the VI classification to hospital-level datasets (including the AHA Annual Survey for hospital characteristics). To identify health plans sponsored by other provider organizations, we additionally search through the full list of MA contracts and flag those whose sponsor is a provider organization. Throughout the descriptive analysis, we classify a hospital as vertically integrated in a given year if it sponsors at least one plan in that year. In the empirical section, we also consider a more robust definition of hospital VI integration status based on the number of beneficiaries enrolled in the hospital-plan relative to the capacity of the hospital.

Readmission measure We use the claims data to compute readmission at the individual level. A patient experiences an acute inpatient readmission if, within 30 days of discharge from the *index* emergency admission, the patient is admitted to any acute-care hospital for any cause. We treat multiple rehospitalizations within the 30-day window as a single event, so the outcome is a readmission probability rather than a count.[3] We apply the transfer exclusion used in the analysis and use admission-type information to exclude admissions identified as planned. This is an approximation to, rather than a complete implementation of, the CMS planned-readmission algorithm.[4] Because our current files are processed separately by calendar year, index admissions near year end may not capture a readmission in the following year. We therefore interpret the displayed MA results as preliminary pending construction of outcomes across adjacent calendar-year files.

Mortality measure We define 30-day mortality as an indicator equal to one when the beneficiary dies during the index hospitalization or within 30 days after discharge. The date of death comes from the MBSF, so the mortality window can extend across calendar years even when the claims files used to construct readmissions are processed separately. Mortality is all-cause and is measured for the same emergency index episodes used in the readmission analysis. Because mortality risk varies substantially across patients, the regressions control for the beneficiary risk score, age, sex, and race in addition to the fixed effects described below.

[3] Gupta (2021) notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

[4] Ziedan and Kaestner (2021) describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

HHI measure We measure hospital market concentration using a 30-minute drive-time market around each hospital. Specifically, for each hospital-year, we define the relevant geographic market as all hospitals located within a 30-minute driving radius and calculate the Herfindahl–Hirschman Index (HHI) as the sum of squared market shares of hospital systems operating within that market. Hospitals belonging to the same system are treated as a single competitor because they are typically jointly owned and may negotiate jointly with insurers. A larger HHI therefore indicates a more concentrated local hospital market. The 30-minute drive-time definition is well suited to hospital competition because hospital markets are inherently local and patients generally view geographically proximate hospitals as the most relevant substitutes. This approach is also closely connected to antitrust analysis of hospital mergers: the DOJ and FTC use the HHI as a standard measure of market concentration, and drive-time-based geographic markets have been used in hospital antitrust analyses and litigation.

We classify markets with an HHI below 1,800 as relatively unconcentrated and markets with an HHI above 1,800 as concentrated. We use the 1,800 threshold because it corresponds to the concentration threshold emphasized in the current Department of Justice and Federal Trade Commission Merger Guidelines, under which an HHI above 1,800 is considered indicative of a highly concentrated market.[5] Our results are not sensitive to this particular threshold and remain robust to alternative HHI cutoffs.

2.5 Summary Statistics

We begin by presenting descriptive statistics for the estimation sample. The estimation sample is restricted to emergency episodes because our instrumental-variable strategy exploits ambulance referral behavior to construct an instrument that addresses patient selection, as described in Section 3.2.

Readmission Rates Table 1 reports average 30-day unplanned readmission rates by hospital integration status and market concentration. In relatively unconcentrated markets, defined as those with an HHI below 1,800, vertically integrated hospitals have a lower readmission rate than non-integrated hospitals, 15.9 percent compared with 16.6 percent. In more concentrated markets, the pattern is reversed: vertically integrated hospitals have a slightly higher readmission rate, 16.3 percent compared with 16.1 percent for non-integrated hospitals. Overall, average readmission rates are very similar across integrated and non-integrated hospitals, at 16.2 and 16.1 percent, respectively.

[5]The 2023 DOJ/FTC Merger Guidelines identify a post-merger HHI above 1,800, together with an increase in HHI of more than 100 points, as giving rise to a structural presumption of competitive harm.

Table 1: 30-Day Unplanned Readmission by Hospital Integration and Market Concentration

Market Concentration	Hospital Integration Non-VI	VI
IV estimation sample		
HHI $<$ 1800	16.6	15.9
HHI $\geq$ 1800	16.1	16.3
Total	16.1	16.2

Notes: Each cell reports the mean 30-day unplanned readmission rate (in percent). Market concentration is measured using 2018 county-level hospital HHI.

Mortality Rates Table 2 reports average 30-day post-discharge mortality rates by hospital integration status and market concentration. In relatively unconcentrated markets, vertically integrated hospitals have a higher raw mortality rate than non-integrated hospitals, 11.0 percent compared with 9.3 percent. In concentrated markets, mortality rates are nearly identical across the two groups, at 10.0 percent for vertically integrated hospitals and 10.1 percent for non-integrated hospitals. Overall, the unconditional mortality rates are also very similar across hospital types. Importantly, these raw differences do not account for patient selection into hospitals. Vertically integrated hospitals may treat patients who differ systematically in underlying health status, severity, or other characteristics that are correlated with mortality. This is particularly relevant in unconcentrated markets, where the higher unconditional mortality rate among vertically integrated hospitals contrasts with our instrumental-variable estimates, which show that admission to a vertically integrated hospital reduces mortality. The difference between the raw and IV estimates is therefore consistent with adverse selection into vertically integrated hospitals and highlights the importance of accounting for endogenous hospital choice.

3 Endogeneity and Identification Strategy

There are two principal endogeneity concerns when estimating the effect of hospital-plan integration on readmission rates. First, hospitals that sponsor plans may differ systematically in unobserved quality from hospitals that do not. Second, patients with different ex ante risks may sort into plan-sponsoring hospitals. This section describes how we address these challenges.

Table 2: 30-Day Post-Discharge Mortality by Hospital Integration and Market Concentration

Market Concentration	Hospital Integration	
	Non-VI	VI
Current IV estimation sample		
HHI < 1800	9.3	11.0
HHI ≥ 1800	10.1	10.0
Total	10.0	10.1

Notes: Each cell reports the mean 30-day post-discharge mortality rate (in percent). Market concentration is measured using 2018 county-level hospital HHI.

3.1 Supply side endogeneity and identification

Hospital side Hospital-by-diagnosis fixed effects absorb time-invariant differences in baseline quality, coding, and discharge practices for each hospital and broad diagnosis group. Year fixed effects absorb common shocks, plan fixed effects absorb persistent differences across MA plans, and beneficiary-residence county fixed effects account for time-invariant differences across patient locations. Residence-county effects are not redundant with hospital effects because the same hospital treats beneficiaries from multiple counties.

Under the standard parallel trends assumption, we identify the hospital-side effect from within-hospital changes in plan-sponsorship status over time, net of aggregate time trends. Table 3 summarizes this variation. A majority of hospitals (64%) are never vertically integrated during the sample period, while 20% remain vertically integrated throughout. Importantly, 16% of hospitals switch status at least once. These "switchers" provide the within-hospital variation used by the fixed-effects specification.

Table 3: Hospital Vertical Integration Status Over Time

	Frequency	Percentage (%)
Always VI	555	19.97
Always non-VI	1,776	63.91
Switchers	448	16.12
Total Hospitals	2,779	100.00

However, patient selection may still pose a concern if the composition of patients changes in response to plan-integration. In particular, if healthier patients differentially sort into vertically integrated settings following changes in integration status, estimates may remain biased. To address this concern, we complement our difference-in-differences strategy with an instrumental variable approach. In section 3.2, we describe instrument we use to account

for patient selection into hospitals.

Another endogeneity concern is that hospitals may adopt VI status as part of a broader organizational change—for example, improvements in care management, discharge planning, or system integration—that begins before formal adoption. In subsection 4.2 we present a validation exercise using hospital-level fee-for-service (FFS) hospital-episode-year data from 2016–2023. The scope of this exercise is to check whether plan-sponsor hospitals and standard hospitals were on different readmission trends prior to the hospital's VI adoption event.

3.2 Instruments for Patient Selection

This section presents the instruments we use to address the possibly non-random selection of beneficiaries between hospitals.

Ambulance-based instrument for hospital assignment In general, patients' choice over hospitals depends unobserved hospital quality which might also be correlated with readmission probability. However, in emergency episodes, the admitting hospital is heavily influenced by pre-hospital factors such as ambulance dispatch, ambulance availability, and patient location. Patients typically do not choose the ambulance company, and dispatch/rotation protocols together with mutual-aid agreements generate plausibly exogenous assignment to ambulance companies with heterogeneous hospital destination preferences (Doyle Jr et al., 2015). The construction is also related to the decision-maker-propensity or "judge leniency" tradition, which uses persistent differences in decision makers' behavior to shift treatment assignment (Dobbie et al., 2018; Bhuller et al., 2020). At the same time, ambulance companies often have stable tendencies to deliver patients to particular hospitals, reflecting historical routing patterns, contracts, operational convenience, or local familiarity.

Figure 1 illustrates the key source of quasi-random variation in hospital assignment. Two ambulances serve the same geographic area but differ in their propensity to transport patients to different hospitals. If one ambulance is systematically more likely to deliver patients to a VI hospital than another, then patients who are quasi-randomly assigned to those ambulances differ in their probability of admission to a VI hospital even though they face similar emergencies and originate from the same local area.

Let H_i be the indicator for *admission to a vertically integrated hospital* for beneficiary-episode i, and let $a(i)$ identify the ambulance company serving episode i for broad diagnosis category $d(i)$. We construct a pooled 2019/2022 leave-one-out ambulance destination

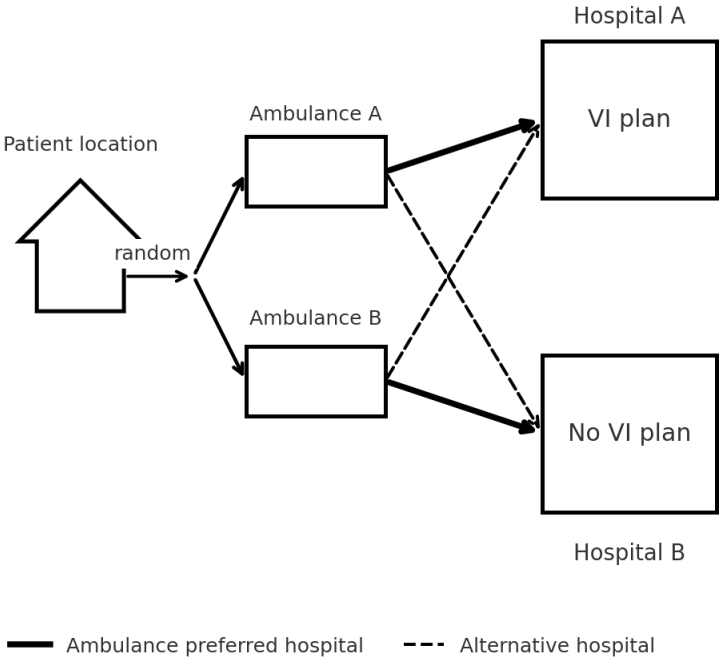


Figure 1: Hospital Assignment and Ambulance Preferences

propensity:

$$Z_i^{A,H} = \frac{1}{N_{a(i),d(i)} - 1} \sum_{\substack{j \neq i: \\ d(j)=d(i)}}^{a(j)=a(i)} H_j.$$

Thus, $Z_i^{A,H}$ is the share of episodes with the same ambulance company and broad diagnosis group that are admitted to a VI hospital, pooling the two sample years and excluding episode i from its own instrument. Conditioning on diagnosis is useful because ambulance routing patterns may differ across emergency conditions. We use an ambulance-by-diagnosis-by-year leave-one-out measure as a robustness specification.

The identifying intuition is that, among otherwise similar patients facing similar emergencies, assignment to an ambulance with a higher propensity to transport patients to VI hospitals raises the probability of admission to a VI hospital while remaining plausibly orthogonal to patient unobservables conditional on location and the control set.

Differential distance We additionally use a geographic shifter based on relative proximity to VI and non-VI hospitals. Let $\text{dist}(r(i), h)$ denote distance from beneficiary i's residence (or pickup location) $r(i)$ to hospital h. Define the nearest VI hospital $h^{VI}(i)$ and the nearest

non-VI hospital $h^{NVI}(i)$, and construct

$$Z_i^{DD} \equiv \text{dist}(r(i), h^{VI}(i)) - \text{dist}(r(i), h^{NVI}(i)).$$

More negative values of Z_i^{DD} indicate greater relative proximity to a VI hospital.

In the hospital-assignment equation, Z_i^{DD} shifts the probability that an emergency episode is delivered to a VI hospital because distance is a first-order determinant of hospital destination in emergency care.

3.3 Excluded instruments

Combining the margins above, the excluded instrument set for admission to a VI hospital is

$$Z_i^H \equiv \left(Z_i^{A,H},\, Z_i^{DD} \right).$$

The first component captures ambulance-specific destination preferences; the second captures relative geographic proximity to VI versus non-VI hospitals. The identifying assumption is that, conditional on the control set and fixed effects, the ambulance preference shifter and differential distance affect readmissions only through admission to a VI hospital. This assumption is most plausible in high-acuity emergency episodes, where patient discretion over the timing and location of care is limited.

A common stability assumption applies to the ambulance-based instrument: conditional on diagnosis, geography, and the fixed effects, ambulance companies' destination preferences do not change in response to hospital plan sponsorship for reasons directly related to post-discharge outcomes. Put differently, plan sponsorship must not induce ambulance providers to re-optimize routing toward the hospital through a channel other than the hospital-assignment margin being instrumented. This assumption is most plausible in high-acuity emergency episodes, where destination patterns are shaped primarily by operational protocols, capacity, and geography.

3.4 Estimation samples

For the hospital-treatment margin, we restrict attention to high-acuity, effectively non-deferrable emergency conditions—such as acute myocardial infarction, stroke, and hip fracture—for which the index admission is largely outside the patient's control. This sharpens the plausibility of the ambulance-based assignment design.

The emergency estimation sample spans a broad set of diagnosis categories. The most common groups are circulatory-system conditions (23.25%), infectious diseases (17.12%), in-

juries and trauma (13.57%), respiratory-system conditions (10.77%), and digestive-system conditions (8.86%). Additional episodes come from genitourinary conditions (6.87%), endocrine disorders (4.48%), and nervous-system conditions (4.00%), with the remaining 11.08% classified as other diagnoses. Thus, while the empirical design is built around emergency and non-deferrable admissions, the estimation sample is not limited to a single clinical condition; rather, it covers a broad range of acute diagnoses for which readmission is a meaningful indicator of inpatient quality and post-discharge care coordination.

4 Evidence from Hospital-Level Traditional Medicare Data

Before turning to the MA beneficiary-level analysis, we present evidence using aggregate hospital-condition outcomes for Traditional Medicare beneficiaries. We do not observe claims-level records for this analysis. The underlying CMS Hospital Compare and HRRP files report risk-standardized readmission and mortality rates as overlapping three-year averages. Adjacent releases therefore share much of the same measurement period and many of the same patients. Treating every overlapping release as a separate time period would create mechanical serial dependence and make event time difficult to interpret. We instead retain four non-overlapping July–June cycles: 2012–2015, 2015–2018, 2018–2021, and 2021–2024.

The resulting panel has far fewer observations than the complete set of overlapping releases, but it spans a substantially longer period than our two years of MA claims. We use that longer panel for two purposes. First, it permits a limited assessment of whether adopting and never-treated hospitals followed different pre-adoption trends. Second, because the outcomes cover Traditional Medicare patients, it tests whether changes associated with MA plan sponsorship extend beyond MA enrollees through hospital-wide changes in care. We implement a staggered difference-in-differences design following Callaway and Sant'Anna (2021).

However, this design has two important limitations. First, aggregate hospital-level data do not allow us to control fully for patient selection across hospitals. Second, the introduction of a sponsored plan may indirectly affect the composition of the observed Traditional Medicare population through switching between Traditional Medicare and Medicare Advantage or through broader changes in referral patterns and system organization. For these reasons, we treat this analysis as complementary to the beneficiary-level analysis in Section 5. A key assumption is that adoption timing is not driven by other contemporaneous organizational changes that independently affect outcomes. We cannot rule out, for example,

that improvements in care management, discharge planning, or system integration coincide with plan sponsorship.

4.1 Staggered Difference-in-differences design

Let h index hospitals, e clinical conditions, and c non-overlapping three-year July–June outcome cycles. Let Y_{hec} denote either the 30-day readmission rate or the risk-standardized 30-day mortality rate for hospital h, condition e, and cycle c. Both outcomes are measured in percentage points rather than logarithms. We estimate the effects of hospital–plan integration separately for each outcome using the staggered difference-in-differences estimator of Callaway and Sant'Anna.

Treatment requires HSA-local hospital–plan integration and the sponsoring system's plans accounting for at least the pooled median share of the HSA Medicare Advantage market, approximately 3.93%. Let G_h denote the first selected outcome cycle with qualifying treatment exposure. Treatment is absorbing after this initial exposure. We use hospital–condition panels covering four non-overlapping three-year cycles and compare treated hospitals with never-treated hospitals.

For each treatment cohort g and outcome cycle $c \geq g$, the estimand is

$$ATT_Y(g, c) = \mathbb{E}[Y_{hec}(g) - Y_{hec}(0) \mid G_h = g], \tag{1}$$

where $Y_{hec}(g)$ denotes the potential outcome under treatment beginning in cycle g, and $Y_{hec}(0)$ denotes the potential outcome in the absence of treatment. Identification relies on conditional parallel trends between each treatment cohort and never-treated hospitals, together with no anticipation of treatment. We aggregate the post-treatment group-time effects into the overall average treatment effect on the treated reported in the results tables.

Estimation uses regression adjustment for hospital ownership, system membership, rural status, region, teaching status, condition indicators, and lagged HSA-wide Medicare Advantage penetration. The specifications do not include the log expected readmission rate. Observations receive importance weights equal to ln(hospital beds), and standard errors are clustered at the hospital level, allowing dependence across conditions and outcome cycles within a hospital.

For each outcome, we estimate the model in the full sample and separately in unconcentrated and concentrated hospital markets. Market concentration is measured using 30-minute-drive hospital HHI, frozen at its 2011 value. Unconcentrated markets have HHI < 1800, while concentrated markets have HHI ≥ 1800. The treatment definition, estimation approach, market classification, weighting, and clustering are common across the two

outcomes; the estimation samples differ because of outcome availability.

Table 4 reports the overall Callaway and Sant'Anna average treatment effects on the treated (CSDID ATTs) for 30-day readmission rates among FFS beneficiaries, using hospital–condition observations over non-overlapping three-year outcome cycles. The dependent variable is measured in percentage points. In the full sample, hospital VI adoption is not associated with a statistically significant reduction in readmissions. We also estimate the effects separately by hospital-market concentration, using 30-minute-drive HHI frozen in 2011. In unconcentrated markets (HHI < 1800), VI adoption is associated with a statistically significant reduction of 0.525 percentage points in the readmission rate. In concentrated markets (HHI ≥ 1800), the estimated reduction is not statistically significant. These results are qualitatively consistent with the hospital-side MA IV results and suggest that reductions in readmissions associated with integration may differ across market environments. They do not identify the causal effect of competition.

Table 4: Impact of Hospital VI Adoption on FFS Readmission Rates

	Full Sample	Unconcentrated Markets	Concentrated Markets
	CSDID ATT	CSDID ATT	CSDID ATT
Hospital VI adoption	-0.0853	-0.5252***	-0.0227
	(0.1044)	(0.1907)	(0.1165)
Observations	33,561	4,273	28,504
Hospital–episode panel	Yes	Yes	Yes
Three-year outcome cycles	Yes	Yes	Yes
Episode indicators	Yes	Yes	Yes
Hospital controls	Yes	Yes	Yes
Lagged HSA MA penetration	Yes	Yes	Yes
Never-treated controls	Yes	Yes	Yes

Notes: The outcome is the 30-day readmission rate in percentage points at the hospital–episode–cycle level, using four non-overlapping three-year July–June cycles. Treatment requires HSA-local hospital–plan integration and sponsored-plan enrollment accounting for at least the pooled median share of the HSA MA market (approximately 3.93%). Treatment begins in the first selected cycle with qualifying exposure and is absorbing. Estimates are overall Callaway and Sant'Anna ATTs, using regression adjustment and never-treated controls. Covariates include hospital ownership, system membership, rural status, region, teaching status, episode indicators, and lagged HSA MA penetration. Concentration uses 30-minute-drive hospital HHI frozen in 2011: unconcentrated markets have HHI < 1800 and concentrated markets have HHI ≥ 1800. Importance weights equal ln(hospital beds). Hospital-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5 shows the corresponding results for mortality under the baseline 30-minute-drive market definition and the 1,800 HHI cutoff. These estimates are not statistically significant.

The beneficiary-level MA analysis below complements these results by addressing patient selection across hospitals with instrumental variables.

Table 5: Impact of Hospital VI Adoption on FFS Mortality Rates

	Full Sample	Unconcentrated Markets	Concentrated Markets
	CSDID ATT	CSDID ATT	CSDID ATT
Hospital VI adoption	-0.0067	0.0929	-0.0221
	(0.0890)	(0.2359)	(0.1007)
Observations	34,543	4,487	29,225
Hospital–condition panel	Yes	Yes	Yes
Three-year outcome cycles	Yes	Yes	Yes
Condition indicators	Yes	Yes	Yes
Hospital controls	Yes	Yes	Yes
Lagged HSA MA penetration	Yes	Yes	Yes
Never-treated controls	Yes	Yes	Yes

Notes: The outcome is the CMS risk-standardized 30-day mortality rate in percentage points at the hospital–condition–cycle level. Conditions include AMI, CABG, COPD, heart failure, pneumonia, and stroke. The analysis uses four non-overlapping three-year July–June cycles, spanning July 2012–June 2024. Treatment requires HSA-local hospital–plan integration and sponsored-plan enrollment accounting for at least the pooled median share of the HSA MA market (approximately 3.93%). Treatment begins in the first selected cycle with qualifying exposure and is absorbing. Estimates are overall Callaway and Sant'Anna ATTs, using regression adjustment and never-treated controls. Covariates include hospital ownership, system membership, rural status, region, teaching status, condition indicators, and lagged HSA MA penetration. Concentration uses 30-minute-drive hospital HHI frozen in 2011: unconcentrated markets have HHI < 1800 and concentrated markets have HHI ≥ 1800. Importance weights equal ln(hospital beds). Hospital-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 Pre-trend assessment

Figures 2 and 3 report dynamic effects from the same Callaway and Sant'Anna estimator used in the main tables. The models use regression adjustment, never-treated hospitals as controls, log-bed importance weights, and hospital-clustered standard errors. Event time is measured in non-overlapping three-year outcome cycles. Event time zero denotes the adoption cycle, while positive event times denote subsequent fully exposed cycles. The figures report point estimates and pointwise 95% confidence intervals; they do not provide a joint test that all pre-treatment coefficients equal zero.

Because the panel contains only four non-overlapping cycles, the event study has at most two reported pre-adoption points and these are available only for cohorts with sufficient pre-treatment history. The exercise is therefore a limited diagnostic, not a strong validation of

parallel trends.

In the unconcentrated-market subsample, the readmission leads are individually statistically indistinguishable from zero, although their confidence intervals are wide. The post-adoption readmission estimates are negative at event times zero through two, with the estimates at zero and two statistically significant. The mortality leads are also statistically indistinguishable from zero, while the post-adoption pattern is mixed. In particular, the negative mortality estimate at event time two is based on only two switching hospitals and should be interpreted cautiously. These event studies provide a diagnostic for pre-adoption trends in the Traditional Medicare sample; they do not directly validate parallel trends among MA beneficiaries.

Appendix A reports the corresponding event-study results for the full sample and concentrated markets. The next section turns to beneficiary-level MA data and uses instrumental variables to address patient selection across hospitals.

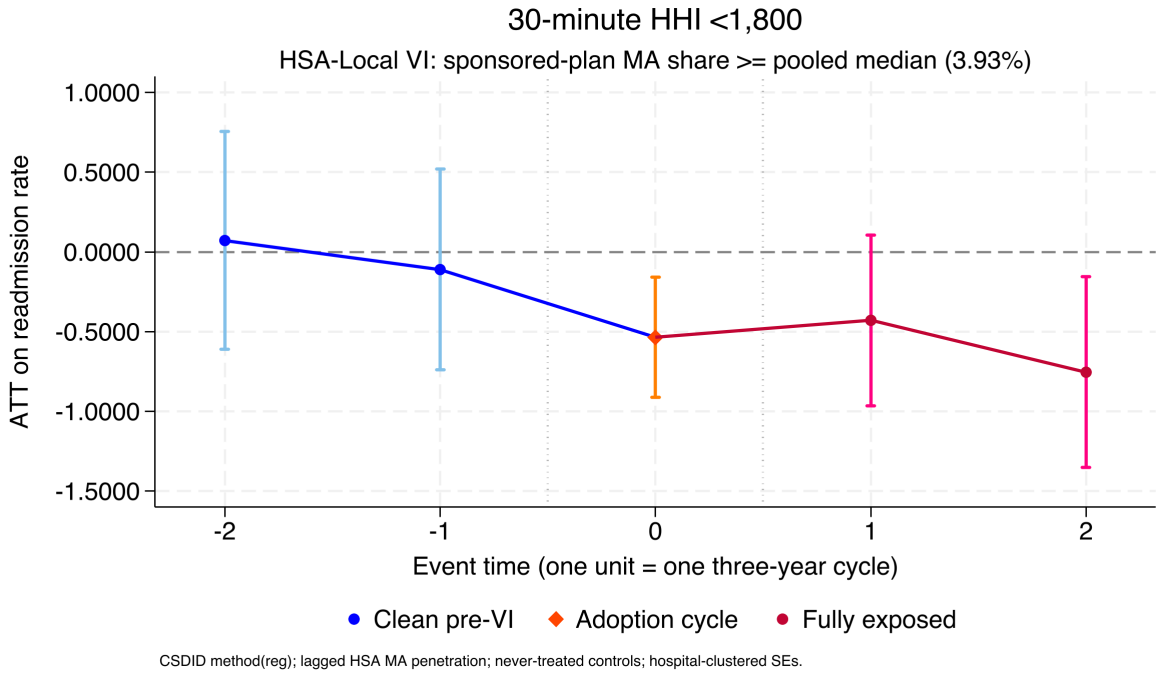


Figure 2: Traditional Medicare readmission event study: Unconcentrated Markets (30-minute HHI below 1,800)

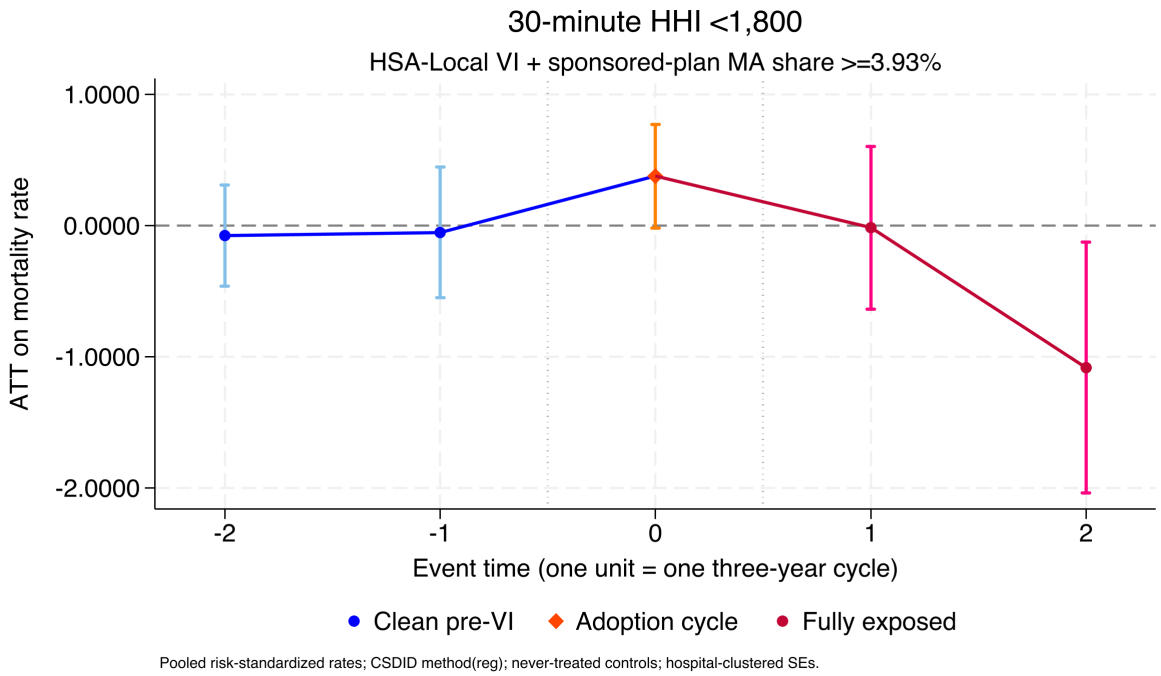


Figure 3: Traditional Medicare mortality event study: Unconcentrated Markets (30-minute HHI below 1,800)

5 Evidence from Beneficiary-Level MA Data

The unit of observation is the beneficiary-by-episode, where an episode is defined as an inpatient admission originating in the emergency department (or an observation stay subsequently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, we observe the beneficiary's Medicare Advantage (MA) plan enrollment, the admitting hospital, and whether the beneficiary experiences an unplanned readmission within 30 days for any cause.

Let i index the beneficiary-episode. The readmission outcome is an indicator for 30-day all-cause unplanned readmission,

$$R_i = \begin{cases} 1 & \text{if beneficiary } i \text{ has an unplanned readmission within 30 days,} \\ 0 & \text{otherwise.} \end{cases}$$

The mortality outcome is

$$M_i = \begin{cases} 1 & \text{if beneficiary } i \text{ dies during the index stay or within 30 days after discharge,} \\ 0 & \text{otherwise.} \end{cases}$$

For the length-of-stay analysis, the outcome is $L_i = \log(\text{inpatient days}_i)$.

Let $h(i)$ denote the admitting hospital, $d(i)$ the broad diagnosis group, $p(i)$ the MA plan, and $t(i)$ the year. The treatment is admission to a vertically integrated hospital:

$$H_i \equiv \mathbb{I}\{\text{beneficiary } i \text{ is admitted to a VI hospital in year } t(i)\}.$$

We report estimates for the full sample and separately for episodes treated in unconcentrated county hospital markets (HHI below 1,800) and concentrated county hospital markets (HHI of at least 1,800). These subsample comparisons do not identify the causal effect of market concentration.

LATE interpretation. The IV estimates identify a local average treatment effect for beneficiaries whose hospital assignment is shifted by ambulance destination patterns and differential distance. Hospital-by-diagnosis fixed effects mean that this complier interpretation is combined with within-hospital changes in VI status across sample years.

5.1 Admission to a VI hospital

We estimate the effect of admission to a vertically integrated hospital on indicators for 30-day all-cause readmission and 30-day mortality. The estimating equation is

$$Y_i = \beta_H H_i + X_i'\gamma + \alpha_{h(i)d(i)} + \phi_{p(i)} + \mu_{c(i)} + \tau_{t(i)} + \varepsilon_i, \tag{2}$$

where X_i contains beneficiary controls, $\alpha_{h(i)d(i)}$ are hospital-by-diagnosis fixed effects, $\phi_{p(i)}$ are plan fixed effects, $\mu_{c(i)}$ are beneficiary-residence county fixed effects, and $\tau_{t(i)}$ are year fixed effects. Residence-county effects are not absorbed by the hospital-by-diagnosis effects: a hospital treats beneficiaries from multiple counties, and beneficiaries residing in a county can be admitted to multiple hospitals. The county effects therefore absorb persistent differences in the populations and local conditions of beneficiaries' places of residence. All MA specifications use the same analytic weights, equal to the product of the log number of ambulance episodes and the log number of hospital admissions. Patient-side selection is addressed by instrumenting H_i with the leave-one-out ambulance destination propensity $Z_i^{A,H}$ and differential distance Z_i^{DD}. Under instrument relevance, the exclusion restrictions, and the assumption that adoption timing is not driven by contemporaneous changes in hospital quality, β_H has a local causal interpretation for beneficiaries whose hospital assignment is shifted by the instruments.

5.2 Results

Readmission Results Table 6 reports the results of the baseline hospital-side specification in Equation 2. The OLS and IV estimates point in very different directions. In the full sample, OLS implies that patients treated at VI hospitals have significantly higher readmission rates, whereas the IV estimate is negative and statistically insignificant. This gap is consistent with, but does not by itself establish, adverse patient selection into VI hospitals. In the unconcentrated-market subsample, the IV coefficient is negative and statistically significant. Conditional on the controls and identifying assumptions, patients admitted to a VI hospital have a 12.44 percentage point lower readmission risk relative to patients admitted to standard hospitals. In the concentrated-market subsample, the IV estimate is negative but statistically insignificant.

Table 6: Readmission: VI Hospital (Baseline Definition)

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
VI Hospital	0.0154***	-0.0259	0.0149*	-0.1244***	0.0134**	-0.0198
	(0.0041)	(0.0166)	(0.0074)	(0.0328)	(0.0049)	(0.0198)
Observations	173,649	173,649	44,366	44,411	126,675	126,675
KP first-stage F		81.103		36.959		69.244
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Residence-county FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital × Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is an indicator for 30-day unplanned readmission. Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with county hospital HHI < 1800; columns (5)–(6) restrict to concentrated markets with county hospital HHI ≥ 1800. All specifications include individual risk score, age, gender, and a white-race indicator, together with year, beneficiary-residence-county, plan, and hospital-by-diagnosis fixed effects. The IV specifications instrument admission to a VI hospital with differential distance and ambulance-based destination preferences. Standard errors, two-way clustered by hospital and broad diagnosis category, are reported in parentheses. The KP statistic is the Kleibergen–Paap rk Wald F statistic. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Length-of-Stay Results Appendix Table 8 reports estimates using log inpatient length of stay. In the full sample, the IV coefficient is -0.103 log points and is statistically significant at the 5% level, corresponding to approximately 9.8% shorter stays. The IV estimates are also negative in both market subsamples but are not statistically significant. These results do not suggest that lower measured readmissions are achieved by keeping patients in the hospital longer, although the subsample estimates are too imprecise to support a stronger conclusion.

Mortality Results Table 7 reports the corresponding hospital-side results for 30-day mortality. In the full sample, both OLS and IV estimates are negative and statistically significant at the 10% level. The IV estimate implies that patients admitted to VI hospitals have a 3.91 percentage point lower mortality risk relative to patients admitted to standard hospitals. Once the sample is restricted to unconcentrated markets, the OLS estimate is negative but statistically insignificant, whereas the IV coefficient is negative and statistically significant at the 5% level. Conditional on the controls and identifying assumptions, patients admitted to

a VI hospital have a 5.62 percentage point lower mortality risk relative to patients admitted to standard hospitals. In concentrated markets, both OLS and IV estimates are negative but statistically insignificant. These findings are qualitatively consistent with the readmission results.

Table 7: 30-Day Mortality: VI Hospital (Baseline Definition)

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
VI Hospital	-0.0090*	-0.0391*	-0.0100	-0.0562**	-0.0073	-0.0266
	(0.0044)	(0.0191)	(0.0103)	(0.0235)	(0.0072)	(0.0198)
Observations	173,649	173,649	44,366	44,366	126,675	126,675
KP first-stage F		81.103		36.959		69.244
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Residence-county FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital × Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is an indicator for 30-day mortality. Columns (1)–(2) use the full estimation sample; columns (3)–(4) restrict to unconcentrated markets with county hospital HHI < 1800; columns (5)–(6) restrict to concentrated markets with county hospital HHI ≥ 1800. All specifications include individual risk score, age, gender, and a white-race indicator, together with year, beneficiary-residence-county, plan, and hospital-by-diagnosis fixed effects. The IV specifications instrument admission to a VI hospital with differential distance and ambulance-based destination preferences. Standard errors, two-way clustered by hospital and broad diagnosis category, are reported in parentheses. The KP statistic is the Kleibergen–Paap rk Wald F statistic. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

5.3 Discussion

The central MA finding is a consistent concentration gradient in the estimated quality effects of hospital plan sponsorship. In unconcentrated markets, admission to a VI hospital is associated with a 12.4 percentage point reduction in 30-day readmission and a 5.6 percentage point reduction in 30-day mortality, and both IV estimates are statistically significant. The corresponding estimates in concentrated markets are smaller and statistically indistinguishable from zero. The full-sample results provide additional evidence of improved performance: although the average readmission effect is imprecisely estimated, the IV estimates indicate significantly lower mortality and shorter length of stay at VI hospitals.

This pattern closely parallels Cooper et al. (2022), who use ambulance assignment to show

that higher-priced hospitals deliver lower mortality only in less concentrated markets. Our results extend that insight from hospital prices to hospital plan sponsorship: the quality gains associated with integration emerge most clearly where hospitals face greater competition. One interpretation is that competitive pressure makes hospitals more likely to translate the internalization of plan-side savings into investments in care management and quality, whereas market power may weaken that discipline.

Taken together, the outcomes are more consistent with improved inpatient and post-discharge performance than with a mechanical reduction in measured readmissions alone. Mortality does not increase, and the length-of-stay estimates do not indicate that hospitals prevent readmissions simply by extending the index stay. Nevertheless, neither outcome identifies the specific organizational mechanism, and the large MA IV magnitudes should be interpreted as local effects for patients whose hospital assignment responds to the instruments rather than as population-wide average effects.

The Traditional Medicare analysis adds a distinct piece of evidence. Its aggregate three-year outcomes cannot address patient selection as flexibly as the MA claims, but its longer panel permits a limited pre-trend assessment and tests for spillovers beyond MA enrollees. The reduction in Traditional Medicare readmissions in unconcentrated markets is consistent with hospital-wide changes in discharge or care management. The baseline Traditional Medicare mortality estimates are imprecise and not statistically significant.

6 Conclusion

This paper studies how hospital–plan integration affects 30-day unplanned readmission and mortality rates. We use a pooled sample of two years of beneficiary-level MA data and implement an instrumental-variables design to address patient selection into hospitals. Differential distance and ambulance-based destination preferences provide variation in admission to a VI hospital. The specifications include plan, beneficiary-residence-county, year, and hospital-by-diagnosis fixed effects and exploit within-hospital changes in VI status over time. In the IV specifications, the full-sample readmission estimate is negative but statistically insignificant, while the mortality estimate is negative and significant at the 10% level; length of stay is also significantly lower. In unconcentrated hospital markets, admission to a plan-sponsoring hospital is associated with a 12.4 percentage point lower readmission probability and a 5.6 percentage point lower 30-day mortality probability relative to admission to a standard hospital. Both estimates are statistically significant. In concentrated markets, neither effect is statistically significant. These patterns suggest heterogeneous effects across market environments.

The Traditional Medicare analysis provides complementary evidence from a longer hospital-level panel. Under the baseline 30-minute-drive market definition, qualifying hospital–plan integration is associated with lower readmission rates in unconcentrated markets, while the mortality estimates are imprecise. The event-study estimates provide no statistically significant evidence of differential pre-adoption trends in the baseline specification. The aggregate data also cannot account for endogenous patient selection as flexibly as the MA design.

Taken together, the findings are consistent with hospital–plan integration changing incentives for care delivery. At a non-integrated hospital, a readmission generates additional inpatient revenue while the patient's MA plan bears the associated cost. At a plan-sponsoring hospital, that conflict remains for patients enrolled in outside plans, but the system also bears downstream spending for patients enrolled in its sponsored plan. Plan sponsorship can therefore make hospital-wide investments that reduce avoidable utilization and improve post-discharge care more valuable. The reduction in MA mortality in unconcentrated markets suggests that lower readmission rates can be accompanied by improvements in patient survival. The corresponding FFS readmission findings are consistent with benefits extending beyond the sponsoring system's MA enrollees. However, these results do not identify the specific organizational changes responsible for the improvements or establish that all patients benefit equally.

These findings contribute to the debate on the potential efficiency gains from hospital–plan integration. Integration can raise concerns about hospital market power, while also creating incentives to reduce avoidable utilization and improve patient outcomes. Our results suggest that effects may vary across market environments, but they do not identify the causal role of competition. The TM and MA analyses provide complementary evidence, but differ in their treatment definitions, market measures, and estimation designs. Moreover, because the MA identification relies on high-acuity emergency admissions, the findings need not generalize to all inpatient care. Future work could examine more discretionary admissions, distinguish hospital-wide changes from benefits specific to sponsored-plan enrollees, and assess effects on post-acute spending and total episode costs.

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A Event Study Estimates and Pre-Trend Tests

This appendix reports event-study estimates corresponding to the baseline difference-in-differences specification using the Callaway and Sant'Anna (2021) estimator. The figures illustrate pre-treatment dynamics and the evolution of outcomes following vertical integration (VI) adoption. The confidence intervals are pointwise and do not constitute a joint test of all pre-treatment coefficients.

The main text presents the unconcentrated-market subsample. Figures 4 and 5 present the corresponding readmission estimates for the concentrated-market subsample and full sample. Figures 6 and 7 present the analogous mortality estimates. The pre-treatment coefficients generally fluctuate around zero, but their confidence intervals are wide. We find no statistically significant overall readmission reduction in the concentrated-market subsample or the full sample.

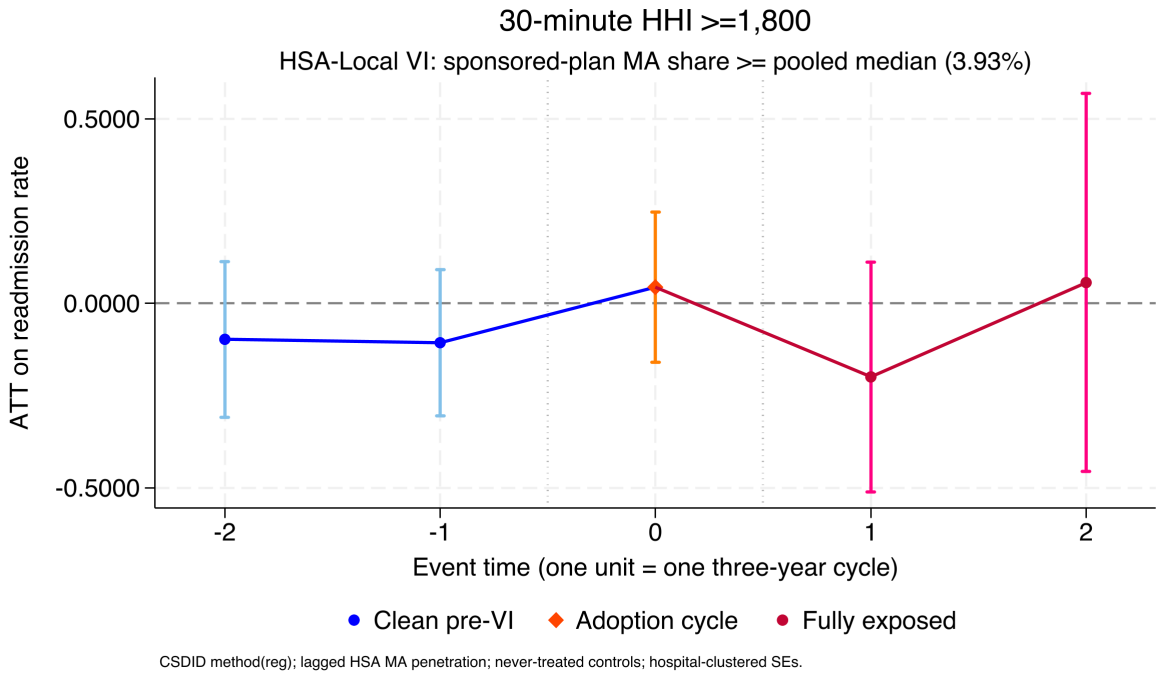


Figure 4: Traditional Medicare readmission event study: Concentrated Markets (30-minute HHI at least 1,800)

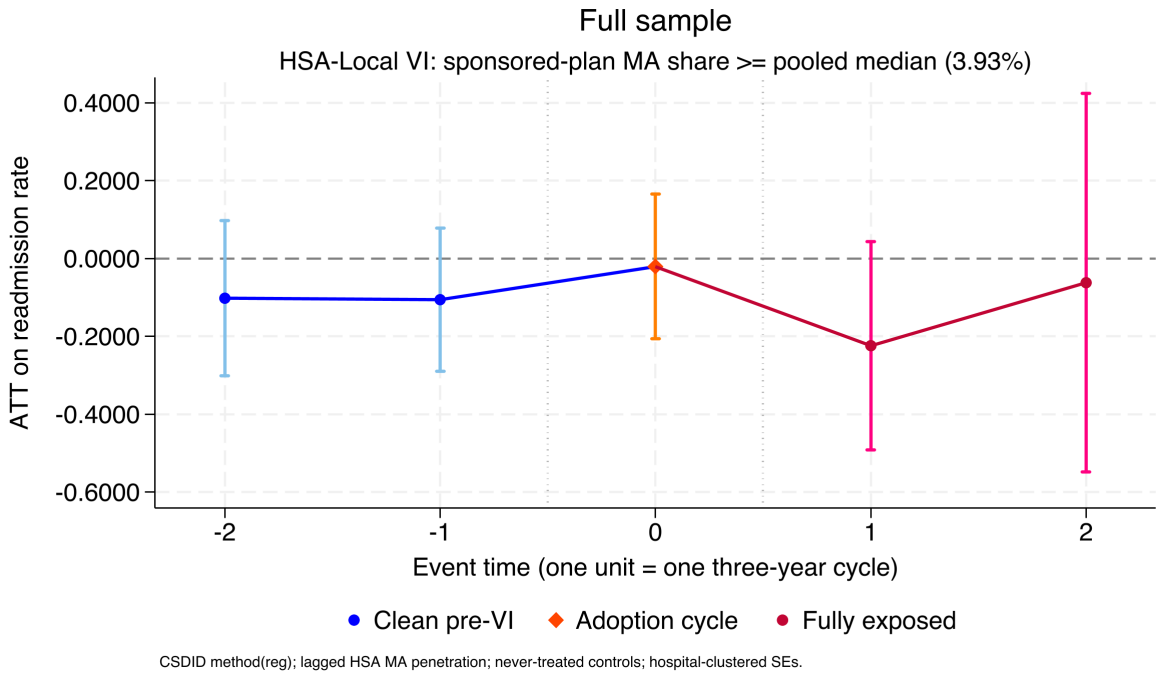


Figure 5: Traditional Medicare readmission event study, full sample

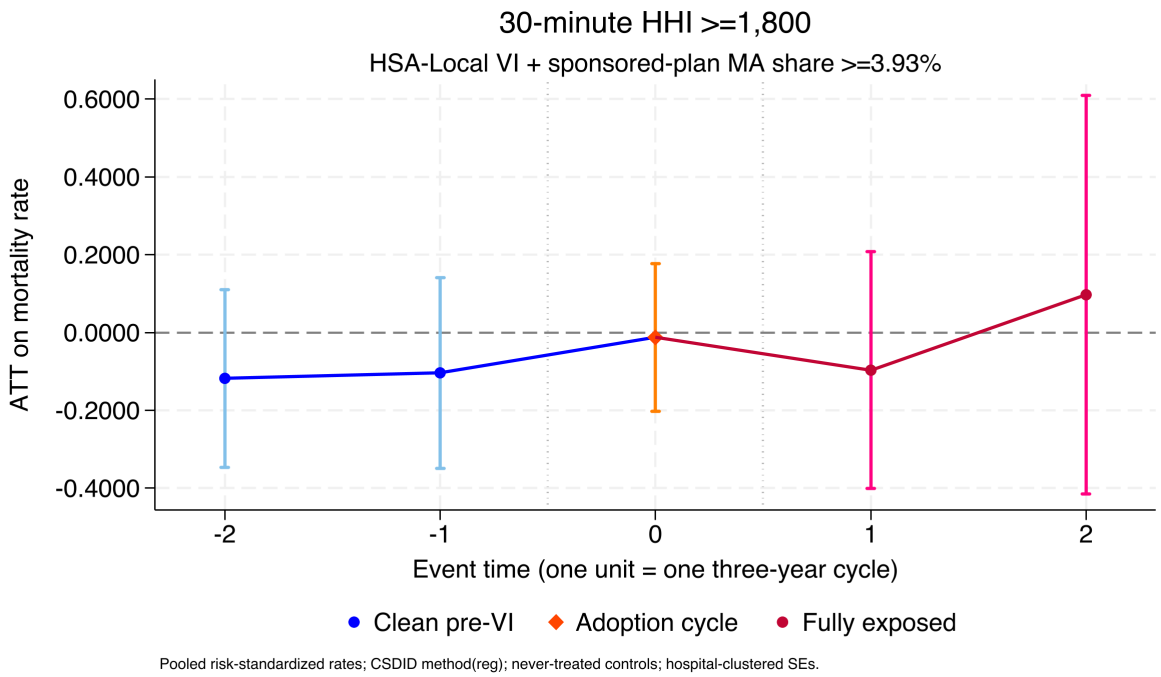


Figure 6: Traditional Medicare mortality event study: Concentrated Markets (30-minute HHI at least 1,800)

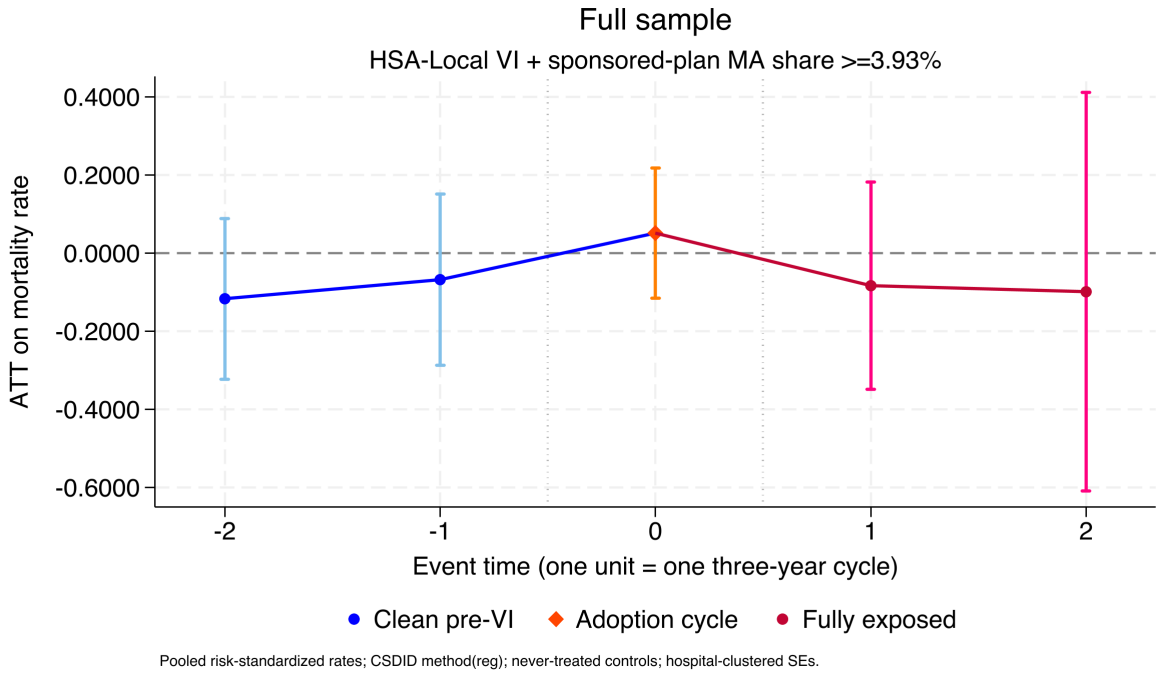


Figure 7: Traditional Medicare mortality event study, full sample

B Additional Medicare Advantage Results

Table 8 reports the Medicare Advantage length-of-stay estimates discussed in Section 5. The outcome is log inpatient length of stay and the treatment, instruments, controls, fixed effects, market split, and clustering follow the corresponding readmission and mortality specifications.

Table 8: Length of Stay: VI Hospital

	Full Sample		Unconcentrated Markets		Concentrated Markets	
	OLS	IV	OLS	IV	OLS	IV
VI Hospital	-0.0126	-0.1032**	-0.0318	-0.0883	-0.0175	-0.0835
	(0.0144)	(0.0380)	(0.0245)	(0.0712)	(0.0174)	(0.0501)
Observations	109,993	109,993	27,466	27,466	80,036	80,036
KP first-stage F		79.473		24.131		69.195
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Residence-county FE	Yes	Yes	Yes	Yes	Yes	Yes
Plan FE	Yes	Yes	Yes	Yes	Yes	Yes
Hospital × Diagnosis FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is log inpatient length of stay. The unconcentrated-market columns restrict the sample to county hospital HHI < 1800; the concentrated-market columns use HHI ≥ 1800. All specifications include the beneficiary risk score, age, sex, and a white-race indicator, together with year, beneficiary-residence-county, plan, and hospital-by-diagnosis fixed effects. The IV specifications instrument admission to a VI hospital with differential distance and ambulance-based destination preferences. Standard errors, two-way clustered by hospital and broad diagnosis category, are reported in parentheses. The KP statistic is the Kleibergen–Paap rk Wald F statistic. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.